NAME: RITHIKA BASKAR **ROLL NO: 231901042**

LABORATORY RECORD

EXP NO.: 7 ENDORSEMENT POLICY CONFIGURATION IN HYPERLEDGER FABRIC

AIM

To configure and implement endorsement policies in Hyperledger Fabric and verify how multiple organizations endorse transactions before they are committed to the blockchain ledger.

PROCEDURE

Step 1: Navigate to the Test Network

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Fabric Network

```
./network.sh up createChannel -ca
```

Step 3: Deploy Chaincode with OR Endorsement Policy

```
./network.sh deployCC -ccl javascript -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript \
  -ccep "OR('Org1MSP.peer','Org2MSP.peer')"
```

Step 4: Verify Chaincode Commitment

```
peer lifecycle chaincode querycommitted -C mychannel
```

Step 5: Initialize the Ledger

```
peer chaincode invoke -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com --tls \
  --cafile $ORDERER_CA -C mychannel -n basic \
  -c '{"function":"InitLedger","Args":[]}'
```

Step 6: Query All Assets

```
peer chaincode query -C mychannel -n basic \
  -c '{"Args":["GetAllAssets"]}'
```

Step 7: Stop the Network

```
./network.sh down
```

Step 8: Restart the Network

```
./network.sh up createChannel -ca
```

Step 9: Deploy Chaincode with AND Endorsement Policy

```
./network.sh deployCC -ccl javascript -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript \
  -ccep "AND('Org1MSP.peer','Org2MSP.peer')"
```

Step 10: Submit a Transaction

```
peer chaincode invoke -o localhost:7050 \   --ordererTLSHostnameOverride orderer.example.com --tls
\
  --cafile $ORDERER_CA -C mychannel -n basic \
  -c '{"function":"CreateAsset","Args":["asset25","Blue","10","Arun","1200"]}'
```

Step 11: Verify the Asset

```
peer chaincode query -C mychannel -n basic \
  -c '{"Args":["ReadAsset","asset25"]}'
```

Step 12: Stop the Network

```
./network.sh down
```

PROGRAM

Endorsement Policy Deployment & Validation Script (deploy_endorsement_policies.sh):

```
#!/bin/bash # Script to configure and test OR and AND Endorsement Policies in Hyperledger Fabric

cd ~/fabric-dev/fabric-samples/test-network

# 1. Start Network and Deploy with OR Endorsement Policy
./network.sh up createChannel -ca ./network.sh deployCC -ccl javascript -ccn basic   -ccp ../asset-transfer-basic/
chaincode-javascript   -ccep "OR('Org1MSP.peer','Org2MSP.peer')"

# Verify Commitment & Initialize Ledger peer lifecycle chaincode
querycommitted -C mychannel
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com   --tls --cafile
$ORDERER_CA -C mychannel -n basic -c '{"function":"InitLedger","Args":[]}'

# 2. Reset Network and Deploy with AND Endorsement Policy
./network.sh down
./network.sh up createChannel -ca ./network.sh deployCC -ccl javascript -ccn basic   -ccp ../asset-transfer-basic/
chaincode-javascript   -ccep "AND('Org1MSP.peer','Org2MSP.peer')"
```

```
# Submit Transaction under AND policy peer chaincode invoke -o localhost:7050 --
ordererTLSHostnameOverride orderer.example.com   --tls --cafile $ORDERER_CA -C mychannel -n basic   -c
'{"function":"CreateAsset","Args":["asset25","Blue","10","Arun","1200"]}'

echo "Endorsement Policies Configured and Executed Successfully."
```

OUTPUT

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network $ ./network.sh up createChannel -
ca
Creating channel 'mychannel'... done
Starting peer0.org1...
Starting peer0.org2...
Starting orderer.example.com...
Network Started Successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ ./network.sh deployCC -ccl
javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript -ccep
"OR('Org1MSP.peer','Org2MSP.peer')"
Packaging Chaincode...
Installing Chaincode...
Approving Chaincode with policy OR('Org1MSP.peer','Org2MSP.peer')...
Committing Chaincode...
Chaincode committed successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network $ peer lifecycle chaincode
querycommitted -C mychannel
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ peer chaincode invoke -o
localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile
$ORDERER_CA -C mychannel -n basic -c '{"function":"InitLedger","Args":[]}'
Ledger Initialized Successfully (Endorsed by Org1)
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network $ ./network.sh down && ./
network.sh up createChannel -ca
Network restarted successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ ./network.sh deployCC -ccl
javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript -ccep
"AND('Org1MSP.peer','Org2MSP.peer')"
Packaging Chaincode...
Installing Chaincode...
Approving Chaincode with policy AND('Org1MSP.peer','Org2MSP.peer')...
Committing Chaincode...
Chaincode committed successfully with AND policy
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network $ peer chaincode invoke -o
localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile
$ORDERER_CA -C mychannel -n basic -c '{"function":"CreateAsset","Args":
["asset25","Blue","10","Arun","1200"]}'
Chaincode invoke successful. status:200
Transaction endorsed successfully by both Org1 and Org2

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ peer chaincode query -C
mychannel -n basic -c '{"Args":["ReadAsset","asset25"]}'
{
  "ID": "asset25",
  "Owner": "Arun",
  "Color": "Blue",
  "Size": 10,
  "AppraisedValue": 1200
}

user@ubuntu:~/fabric-dev/fabric-samples/test-network $ ./network.sh down
Network cleaned up successfully
```

RESULT

The endorsement policies were successfully configured in the Hyperledger Fabric network. Transactions were validated according to the specified endorsement rules (`OR` and `AND` policy configurations), demonstrating how Hyperledger Fabric ensures trust, integrity, and secure transaction processing through endorsement policies.